

Manufacturer	Customer	Business Transaction	
ACME Metal Works GmbH Industrial Park 123 52066 Aachen DE quality@acme-metal.example.com	European Research Institute Science Boulevard 42 75015 Paris FR materials@eri.example.org	Order	PO-23456 · 1000 kg Pos. 1 · 2025-05-01
		Delivery	DN-65432 · 1000 kg Pos. 1 · 2025-05-16

Product			
Product Name	Titanium Alloy Ti-6Al-4V ELI	Number (UNS)	R56401
Batch ID	T-65432-01	Shape	RoundBar - D 30 mm - Length 3000 mm
Surface Condition	Machined	Product Norm	ASTM F136 (2022)
Weight	1000 kg		
Production Date	2025-05-15		
Country of Origin	DE		

Chemical Analysis								
Heat Number T-65432 - Melting Process VAR - Casting Method VacuumCasting - Casting Date 2025-05-14 - Sample Location Product								
Symbol	Ti	Al	V	Fe	O	C	N	H
Unit	%	%	%	%	%	%	%	ppm
Min		5.50	3.50					
Max		6.50	4.50	0.25	0.13	0.080	0.050	120
Actual	89.32	6.02 ± 0.05	3.95 ± 0.03	0.18	0.11	0.026	0.012	35

Mechanical Properties						
Property	Actual		Minimum	Maximum	Method	Status
Fatigue Test					ASTM E466	✓
Room temperature, R = 0.1						
Cycles (N)	10000	50000	100000	500000	1000000	10000000
Value	650	635	610	590	570	550
Notch Sensitivity	0.85 - 0.92				Internal Method TS-5432	✓

Property	Actual	Target/Min	Maximum	Method	Status
Density	4.43 ± 0.01 g/cm³	4.43 g/cm³		ASTM B311	✓

Supplementary Tests								
Test	Result / limits					Method	Status	
Microstructure Examination	Equiaxed alpha with intergranular beta					ASTM E407	✓	
Ultrasonic Inspection	Yes - No indications greater than reference standard					ASTM E2375	✓	
Surface Quality Assessment	Class 1 - Medical Grade					Visual Inspection per ASTM F136	✓	
Alpha Case Depth	5 µm - max 25 µm					Microhardness Traverse	✓	
Grain Size Distribution	8 - 10 ASTM No. - target 7 - 12 ASTM No.					ASTM E112	✓	
Hardness Profile (ASTM E384), values in HV								
Distance from surface (mm)	0.1	0.5	1	2	3	5	10	15
Value [HV]	345	350	352	350	349	348	351	347

Validation
We hereby certify that the material described above has been manufactured and tested in accordance with ASTM F136 and meets all requirements for surgical implant applications.
Validated by Dr. Markus Weber, Head of Metallurgy, Research & Quality - 2025-05-17